

Dr Subash Sharma

Materials Scientist | Thin-Film Growth and Electron Microscopy

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PROFESSIONAL PROFILE

Materials scientist with over 12 years of research experience in thermal CVD, two-dimensional materials and electron microscopy. Experience spans graphene, hBN and transition-metal dichalcogenide synthesis, a Graphene Flagship hBN collaboration between Cambridge and AIXTRON, and industrial TaC coating development. Combines thin-film growth and characterisation with in-situ studies of materials transformations. Developing Python-based microscopy analysis tools and workflows for AI-assisted materials discovery.

CORE TECHNICAL EXPERTISE

Thin-film synthesis: Hands-on thermal CVD, including atmospheric- and low-pressure processes; graphene, hBN and TMD synthesis; sputtering; thin-film transfer and growth-process optimisation.

Complementary deposition knowledge: MOCVD and ALD process principles; involvement in MOCVD equipment procurement at the Cambridge Graphene Centre.

Characterisation: TEM/STEM, HRTEM, HAADF, EELS, EDX and SAED; in-situ heating and electrical biasing; SEM/EDX, Raman spectroscopy, XPS and AFM.

Computing and laboratory practice: Python, NumPy, pandas and Streamlit; microscopy data analysis; vacuum systems, instrument troubleshooting, user training, SOPs and laboratory safety.

RESEARCH AND PROFESSIONAL EXPERIENCE

Technical Specialist in Electron Microscopy

July 2024–present

University of Bristol, School of Chemistry

- Provide electron microscopy support for materials research, combining TEM/STEM and SEM/EDX characterisation with interpretation of structure and composition.
- Oversee instrument operation, troubleshooting, maintenance coordination and user training; develop operating procedures and competency assessments.
- Develop Python-based diffraction analysis tools and a materials-discovery workflow linking predicted structures with crystallographic databases and literature.

Research Scientist

December 2023–June 2024

Advanced Furnace Technology, Cambridge

- Developed TaC coating processes for high-temperature industrial applications.
- Worked on furnace process scale-up, including temperature profiles, gas flow and deposition parameters, with attention to repeatability and manufacturing requirements.

RESEARCH AND PROFESSIONAL EXPERIENCE CONTINUED

Postdoctoral Researcher

December 2019–November 2023

Cambridge Graphene Centre, University of Cambridge

- Responsible for Cambridge's contribution to a Graphene Flagship project on hBN growth on 200 mm wafers, in collaboration with AIXTRON.
- Contributed to thin-film characterisation using Raman spectroscopy, AFM, SEM, XPS and electron microscopy to assess film structure and quality.
- Developed polymer-assisted transfer processes for integrating two-dimensional materials onto other substrates.
- Contributed to collaborative project deliverables, coordination with European partners and student supervision in materials characterisation and analysis.
- Participated in procurement of an MOCVD system for the Cambridge Graphene Centre, drawing on knowledge of deposition processes and research requirements.

Postdoctoral Fellow

April 2015–November 2019

Nagoya Institute of Technology, Japan

- Investigated thermal CVD growth of graphene, hBN and transition-metal dichalcogenides, including the effects of precursors, substrates, gas composition and growth conditions.
- Studied morphology-controlled hBN synthesis and CVD growth of WS₂ structures, linking processing conditions with resulting material morphology.
- Used in-situ TEM with electrical biasing and heating to investigate catalyst-mediated graphitisation, graphene formation and carbon nanotube transformations.
- Investigated metal–carbon interactions across multiple catalyst systems, including Cu, Mo, Sn, In, Ga, Co and Pd.
- Operated CVD, sputtering and microscopy equipment, and mentored undergraduate and postgraduate researchers.

SELECTED COMPUTING AND MATERIALS DISCOVERY PROJECTS

Crystal-structure triage and literature cross-referencing

Developing SSB-CrossRef to compare predicted inorganic crystal structures with the Crystallography Open Database, Materials Project and published literature, supporting assessment of prior experimental evidence and candidate novelty.

Electron diffraction analysis

Developing a Python and Streamlit application for SAED ring detection and diffraction-data analysis, including DM3 and DM4 microscopy files.

Awarded research computing allocation

Isambard 3: Generative AI for Inorganic Materials Discovery and Triage. Awarded 1,000 node-hours in 2026; project runs to June 2027.

PROFESSIONAL SERVICE

Researcher training and mentoring; peer review for Nano Letters, Nanoscale and Nanotechnology. Outstanding Reviewer Award, Nanotechnology (IOP, 2017).

EDUCATION

PhD in Engineering | Nagoya Institute of Technology, Japan | 2015

CVD synthesis of graphene and hBN using solid precursors

Master of Engineering | Chubu University, Japan | 2012

TiN/TiO₂ film deposition and properties

BSc in Physics | Tribhuvan University, Nepal | 2008

SELECTED PUBLICATIONS AND PREPRINT

2026 | Identification of Cu-N₂ Sites for Zn-Air Batteries in Harsh Electrolytes: Computer Virtual Screening, Machine Learning, and Practical Application

C. Xu, K. Li, S. Liu, J. Xu, S. Sharma et al.

CCS Chemistry 8(3), 1607–1621. [Publication ↗](#)

2025 | Graphene Electro-Absorption Modulators for Energy-Efficient and High-Speed Optical Transceivers

M. Tiberi, A. Montanaro, C. Wen, J. Zhang, O. Balci, S. M. Shinde, S. Sharma et al.

arXiv:2506.03281. Preprint. [Preprint ↗](#)

2025 | In Situ TEM Study of Electrical Property and Mechanical Deformation in MoS₂/Graphene Heterostructures

S. Giri, S. Sharma, R. D. Mahyavanshi, G. Kalita, Y. Yang and M. Tanemura.

Nanomaterials 15(2), 114. [Publication ↗](#)

2023 | Near- and Far-Field Observation of Phonon Polaritons in Wafer-Scale Multilayer Hexagonal Boron Nitride Prepared by Chemical Vapor Deposition

E. Calandrini et al. (including S. Sharma).

Advanced Materials 35, 2302045. [Publication ↗](#)

2018 | Synthesis of Freestanding WS₂ Trees and Fibers on Au by Chemical Vapor Deposition (CVD)

S. Sharma et al.

physica status solidi (a) 215, 1700566.

2017 | Optimization of CVD parameters for graphene synthesis through design of experiments

R. Papon, C. Pierlot, S. Sharma et al.

physica status solidi (b) 254, 1600629.

2016 | Morphology-controlled synthesis of hexagonal boron nitride crystals by chemical vapor deposition

S. Sharma et al.

Crystal Growth & Design 16, 6440–6445.

2014 | Synthesis of graphene crystals from solid waste plastic by chemical vapor deposition

S. Sharma et al.

Carbon 72, 66–73.

Full publication list available through Google Scholar.